1.1. The term Measurement and Methods of Measurement

Definition and concept of Disasters

- "**Disaster Management** can be defined as the organization and management of resources and responsibilities for dealing with all humanitarian aspects of emergences, in particular preparedness, response and recovery in order to lessen the impact of disasters".
- The **United Nations** defines a disaster as a serious disruption of the functioning of a community or a society.
- Disasters involve widespread human, material, economic or environmental impacts, which exceed the ability of the affected community or society to cope using its own resources.
- The **World Health Organization** defines Disaster as "any occurrence that causes damage, ecological disruption, loss of human life, deterioration of health and health services, on a scale sufficient to warrant an extraordinary response from outside the affected community or area".

1.2. Hazard

Concept

- A **hazard** is a situation that poses a level of threat to life, health, property, or environment.
- A dangerous phenomenon, substance, human activity or condition that may cause loss of life, injury or other health impacts, property damage, loss of livelihoods and services, social and economic disruption, or environmental damage hazards are conditions that have the potential to harm to a community or environment.

Table 1 Examples of Hazards and their Effects

Table: Examples of Hazards and their Effects		
Workplace hazard	**Example of hazard**	**Example of harm caused**
Source of energy	Electricity	Shock, electrocution
Condition	Wet floor	Slips, falls
Thing	Knife	Cut
Substance	Benzene	Leukemia
Practice	Hard rock mining	Silicosis
Process	Welding	Metal fume fever

- As shown from previous table, Workplace hazards also include practices or conditions that release uncontrolled energy like:
  - An object that could fall from a height (potential or gravitational energy),
  - A run-away chemical reaction (chemical energy),
  - The release of compressed gas or steam (pressure, high temperature),
  - Entanglement of hair or clothing in rotating equipment (kinetic energy),

- Contact with electrodes of a battery or capacitor (electrical energy).

- So, hazards can include conditions, objects and practices which can threaten or tend to threatened life or property.

Types of Hazard

- A common way to classify hazards is by category.

1) Biological Hazard
   - Bacteria, viruses, insects, plants, birds, animals, and humans, etc.
2) Chemical Hazard
   - Depends on the physical, chemical and toxic properties of the chemical.
3) Ergonomic Hazard
   - Repetitive movements, improper set up of workstation, etc.
4) Physical Hazard
   - Radiation, magnetic fields, high pressures, noise, etc.
5) Psychosocial Hazard
   - Stress, violence, etc.
6) Biological Hazard
   - Slipping/tripping, equipment malfunctions or breakdowns, etc.

1.3. Vulnerability

Concept

- In relation to hazards and disasters, vulnerability is a concept that links the relationship that people have with their environment to social forces and institutions and the cultural values that sustain and contest them.
- Exposed to the possibility of being attacked or harmed, either physically or emotionally.
- Vulnerability refers to the inability to withstand the effects of a hostile environment.

Figure 1 Importance of Vulnerability

1.4. Risk

Concept

- Risk is the potential of losing something of value.
- The values such as physical health, social status, emotional well-being or financial wealth.
- A probability or threat of damage, injury, liability, loss, or any other negative occurrence that is caused by external or internal vulnerabilities.

- Risk is an uncertain event or condition that, if it occurs, has an effect on at least one objective.
- Risk means 'effect of uncertainty on objectives'.
- So, Risk is the chance or probability that a person will be harmed or experience an adverse health effect if exposed to a hazard. It may also apply to situations with property or equipment loss.
- Factors that influence the degree of Risk include:
  - How much a person is exposed to a hazardous thing or condition?
  - How the person is exposed, e.g. breathing in a vapour, skin contact etc.
  - How severe are the effects under the conditions of exposure.
- What is Risk Assessment?

Figure 2 Chart of Risk Management Process

- Risk Assessment is the process where you:
  - Identify hazards,
  - Analyse or evaluate the risk associated with that hazard,
  - Determine appropriate ways to eliminate or control the hazard, and
  - Review the ways of elimination or control.

1.5. Capacity

Concept

- Capacity is:
  - The ability to receive, hold or absorb something.
  - The ability to do, make, or accomplish something.
  - The ability or power to contain, absorb, hold, or do some work to defend.
- Emergency
  - It is a deviation from planned or expected behaviour or course of events that endangers or severely affect people and property and environment.

1.6. Disaster Management

Concept

- Disaster Management is the planned steps taken to minimize the effects of a disaster, and to be able to proceed to business continuity stage.

- Disaster Management includes sum total of all activities, programme and measures which can be taken up before, during and after a disaster with the purpose of avoiding, reducing the impact or recovering from its losses.

Figure 3 various types of disasters

- Disaster Management means managing resources and various responsibilities to deal with all humanitarian aspects or emergencies.
- This may include preparedness before disaster, response and recovery i.e. rebuilding and supporting society.
- The purpose of this is to lessen the impact of disasters.

Aims of Disaster Management

- The aims of Disaster Management are:
  - To reduce the potential losses from hazards (avoid, if possible).
  - To assure prompt and appropriate assistance to victims when necessary.
  - To achieve rapid and durable recovery.

Importance of Disaster Management

- Over past 20 years disasters have affected 4.4 billion people, caused $2 trillion of damage and killed 1.3 million people.
- These losses have outstripped the total value of official development assistance in the same period.
- Natural disasters disproportionately affect people living in developing countries and the most vulnerable communities within those countries.
- At the global level, there has been considerable concern over natural disasters.
- Even as scientific and material progress is made, the loss of lives and property due to disasters has not decision.
- India has been traditionally vulnerable to natural disasters on account of its unique geo-climate conditions.
- Floods, droughts, cyclones, earthquakes and landslides have been recurrent phenomena.

Phases of Disaster Management

- Disaster Management activities can be grouped into five phases that are related by time and function to all types of emergencies and disasters.
- These phases are also related to each other, and each involves different types of skills.
- These phases are:
  - Planning and Mitigation
  - Preparedness
  - Response
  - Recovery

Figure 4 Classification of phases of disaster management

Planning Phase

- Activities necessary to analyse and document, the possibility of an emergency or disaster and the potential consequences or impacts on life, property, and the environment.
- This includes assessing the hazards, risks, mitigation, preparedness, response, and recovery needs.

Mitigation Phase

- Activities that actually eliminate or reduce the probability of a disaster (e.g. arms build-up to deter enemy attack).
- It also includes long-term activities designed to reduce the effects of unavoidable disaster (e.g. land use management, establishing comprehensive emergency management programs such as vegetation, building restrictions in potential flood zones).

Preparedness Phase

- Activities necessary to the extent that mitigation measures have not, or cannot, prevent disasters.
- In the preparedness phase, governments, organizations, and individuals develop plans to save lives and minimize disaster damage (e.g. compiling state resource inventories, installing early warning systems, and preparing predetermined emergency response forces).

- Preparedness measures also seek to enhance disaster response operations (e.g. by stockpiling vital food and medical supplies, through training exercises and by mobilizing emergency response personnel on standby).

Response Phase

- Activities following an emergency or disaster.
- These activities are designed to provide emergency assistance for victims (e.g. search and rescue, emergency shelter, medical care, and mass feeding).
- They also seek to stabilize the situation and reduce the probability of secondary damage (e.g. shutting off contaminated water supply sources, and securing and patrolling areas prone to looting) and to speed recovery operations (e.g. damage assessment).

Recovery Phase

- Activities necessary to return all systems to normal or better.
- They include two sets of activities:
  - Short-term recovery activities return vital life-support systems to minimum operating standards (e.g. clean-up, temporary housing, and access to food and water).
  - Long-term recovery activities may continue for a number of years after a disaster. Their purpose is to return life to normal or improved levels (e.g. redevelopment loans, legal assistance, and community planning).

1.7. Foundation of Emergency Management

- All phases of emergency management depend on data from a variety of sources.
- The appropriate data has to be gathered, organized, and displayed logically to determine the size and scope of emergency management programs.
- During an actual emergency it is critical to have the right data, at the right time, displayed logically, to respond and take appropriate action. Emergencies can impact all or a number of government departments.
- Emergency personnel often need detailed information concerning pipelines, building layout, electrical distribution, sewer systems and so forth.
- By utilizing a GIS, all departments can share information through database on computer-generated maps in one location.
- Most emergencies do not allow time to gather these resources. This results in emergency responders having to guess, estimate, or make decisions without adequate information.
- Most of the data requirements for emergency management are of a spatial nature and can be located on a map.